Readme:

Max Number of Robots and Sites:

- #Number of Robots : **N**
- #Number of Sites : **M**

=> the Equation:

$$$(7 * N+ N * M + 6 * M < 18275)$$$

google sheet max available number of columns?

As of **2025**, Google Sheets supports up to:

- 🧮 **18,278 columns** (i.e., from column **A** to **ZZZ**).
- 📄 **10 million cells total** per spreadsheet (this limit includes all sheets in the file).

So if you have many rows, your maximum number of columns may be limited by that **10 million cell** total.

Example:

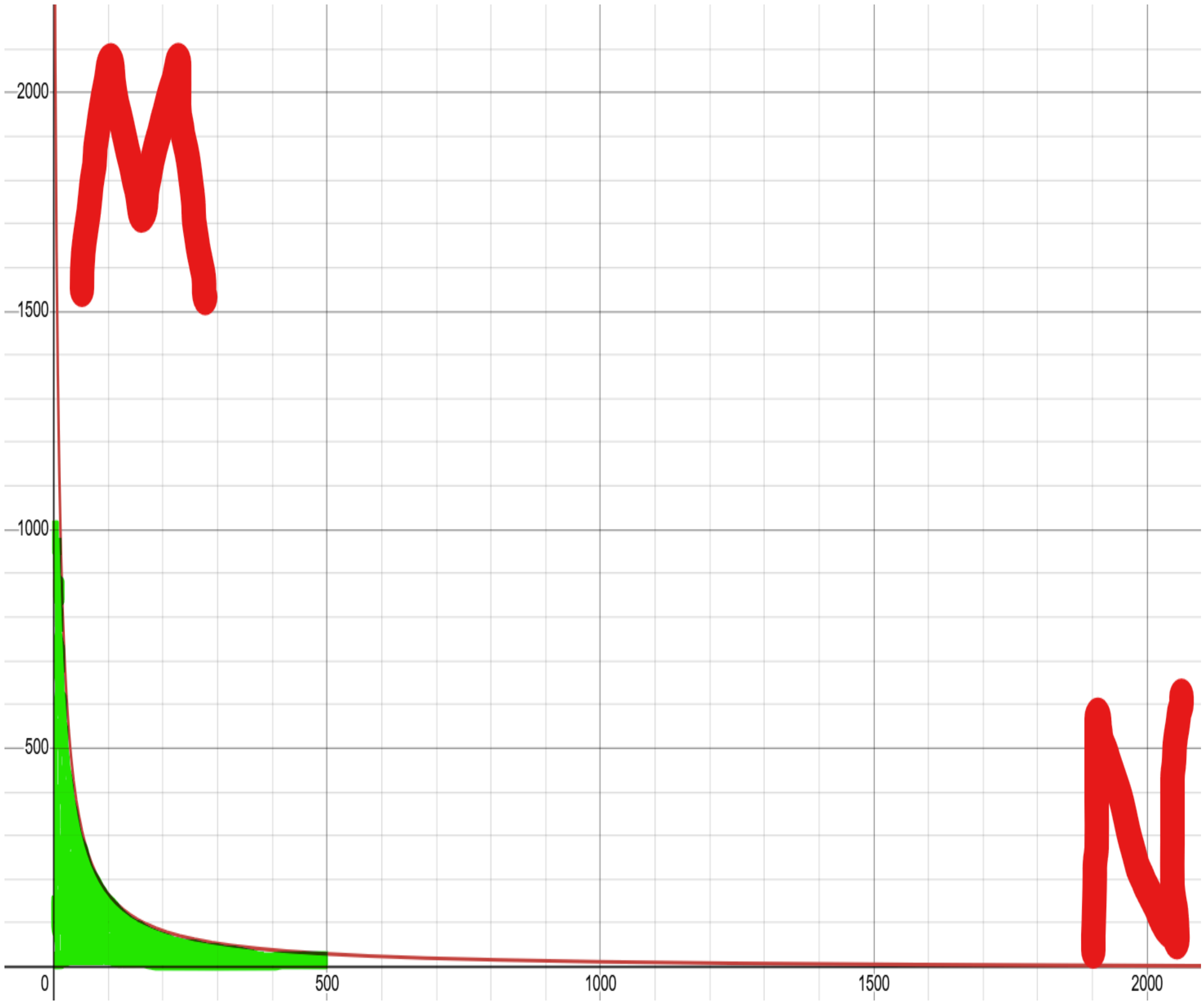
If you use 1,000 rows, the max columns you can have is:

10,000,000 ÷ 1,000 = **10,000 columns**

But the absolute upper cap is still **18,278 columns** — you can't exceed that even if you have few rows.

Would you like me to show how to check how many columns your current sheet is using or how close it is to the limit?

- So the boundaries is:



- So the curve passes roughly through points:
- (N=1 Robot, M=2610 Site)
- (N=2284 Robot, M=1 Site)

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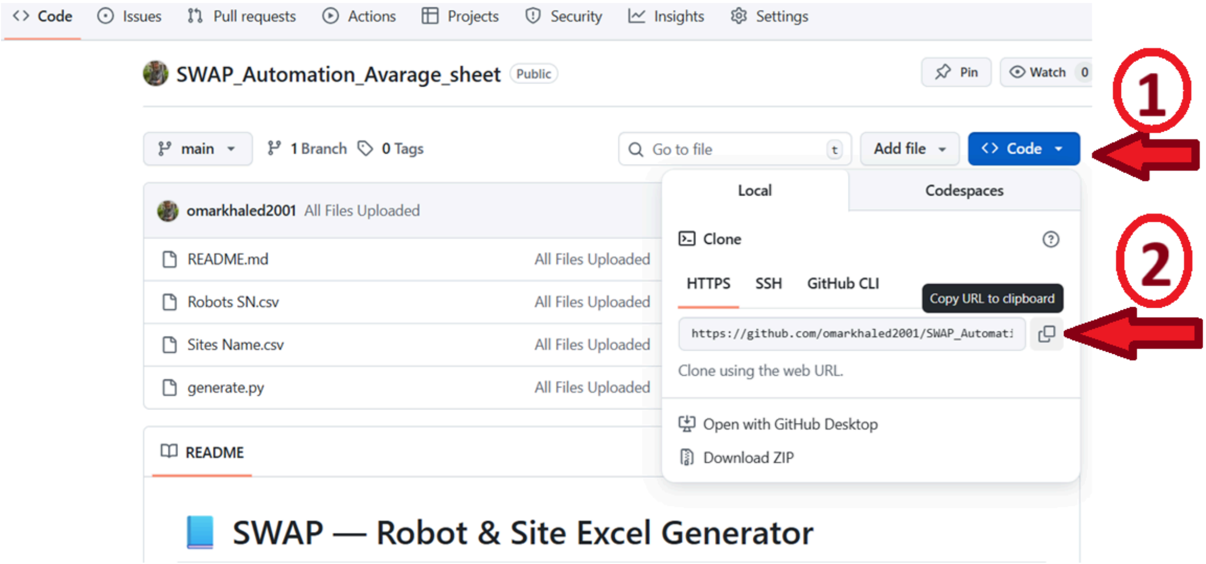
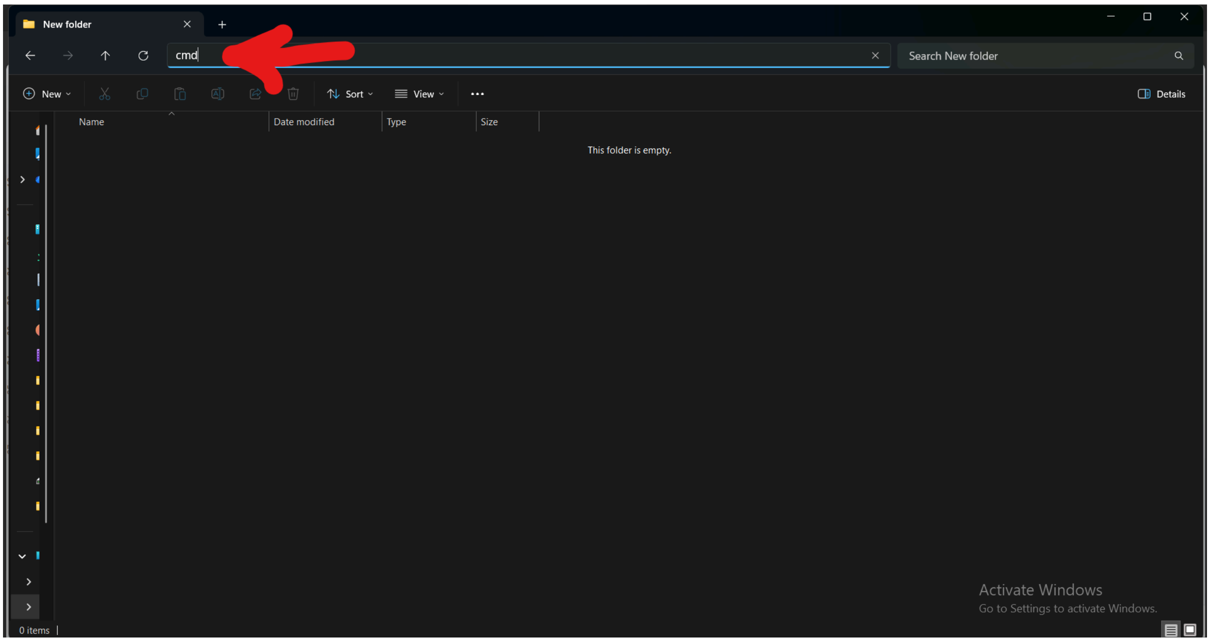
How to Use:

Step 1:

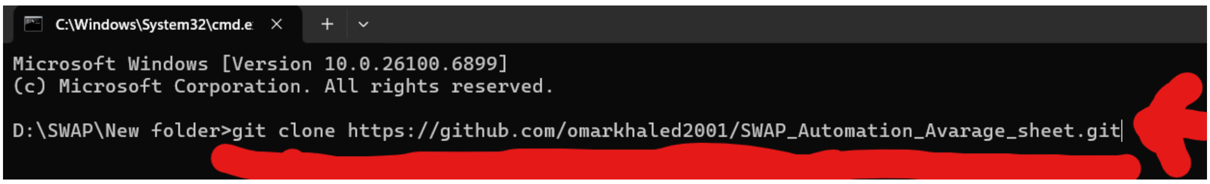
- Go to this Link: https://github.com/omarkhaled2001/SWAP_Automation_Avarage_sheet.git
- To Download the project follow this instructions:
  - First:
    - First:
      Copy URL of the project to download it



- Second:
  Create New Folder - then write "cmd" in this bar



- Third:
  Write this command "git clone " then paste the Project URL and then "Enter"



- You will find 4 files:
  - `generate.py`

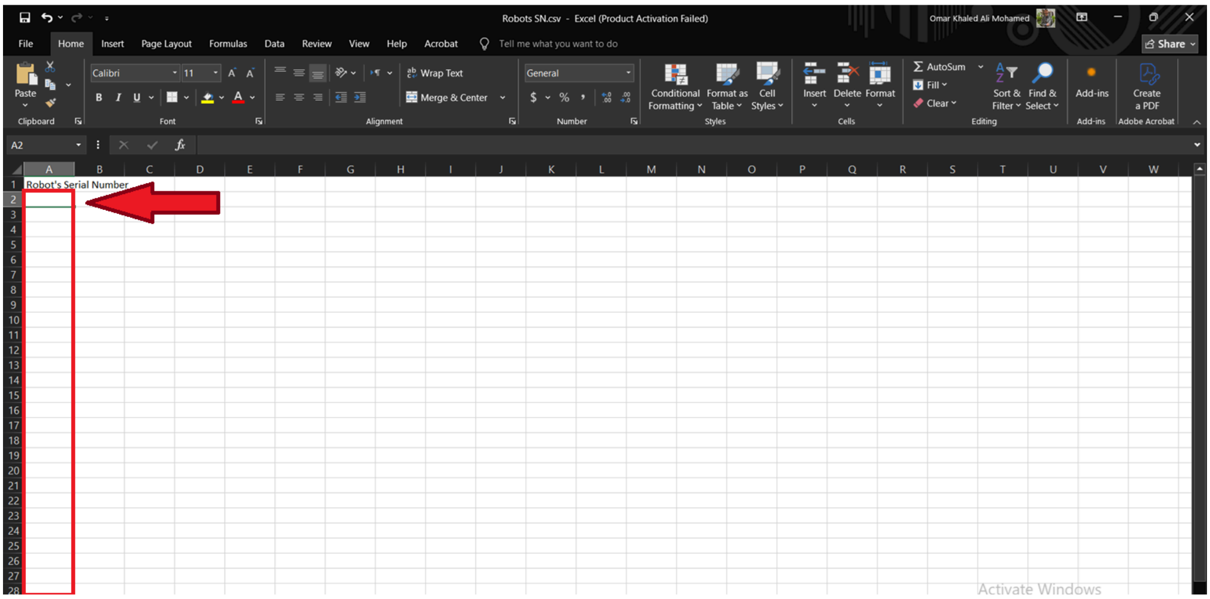
- `Robots SN.csv`
- `Sites Name.csv`
- `Readme.md`

Step 2:

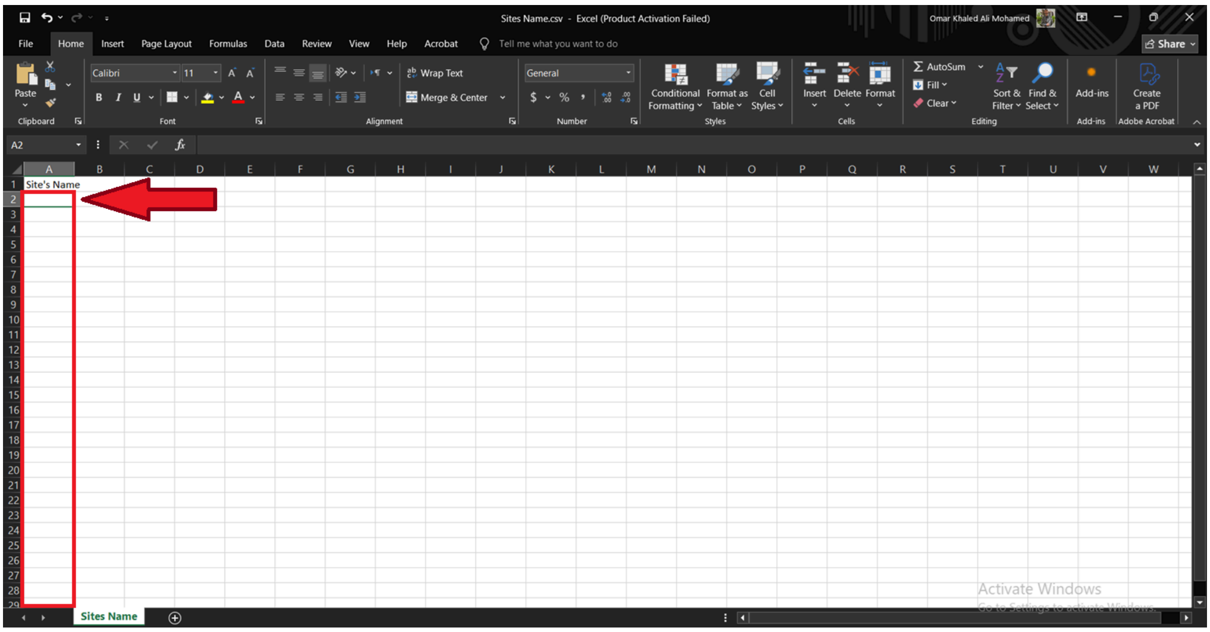
- Read Carefully instructions in:
  - `Readme.md file`
  - `How to Use pdf file`

Step 3:

- Fill `Robots SN.csv` with your Robots Serial Number.

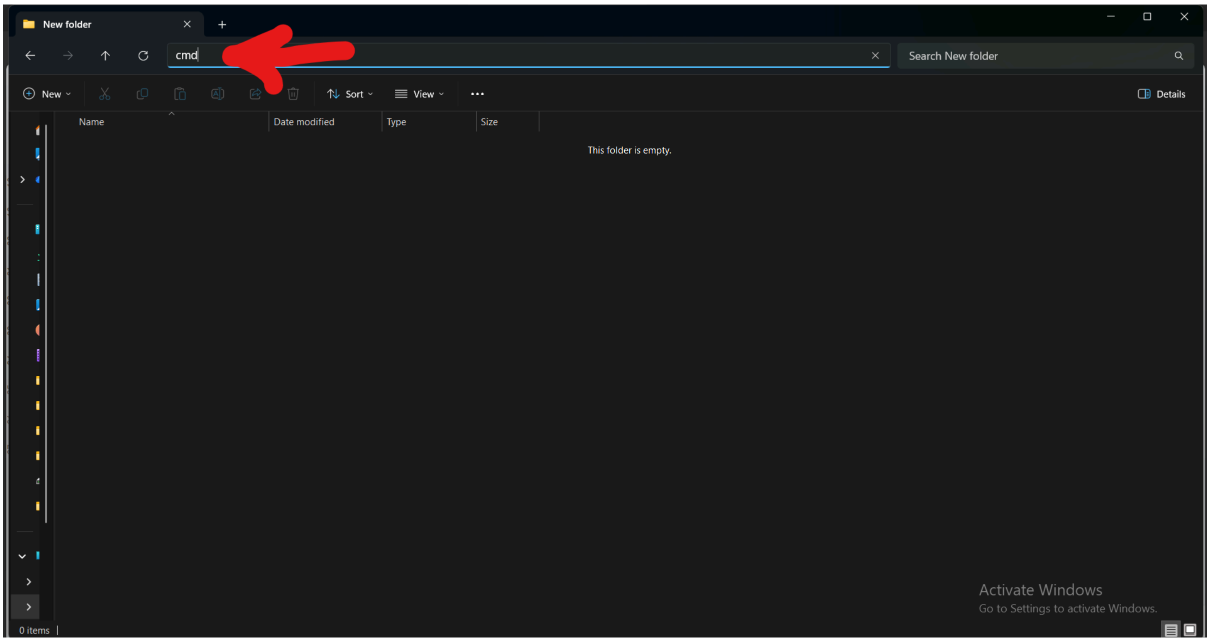


- Fill `Sites Name.csv` with your Sites Name.

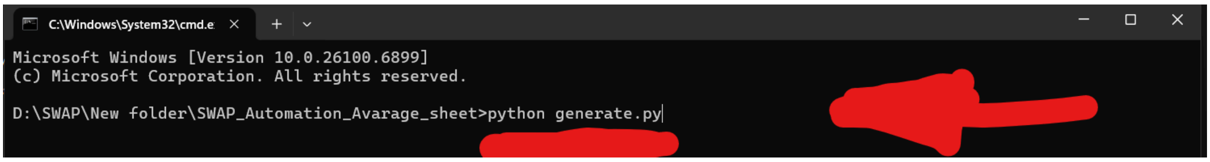


Step 4:

- In the same folder write “cmd” in this bar.

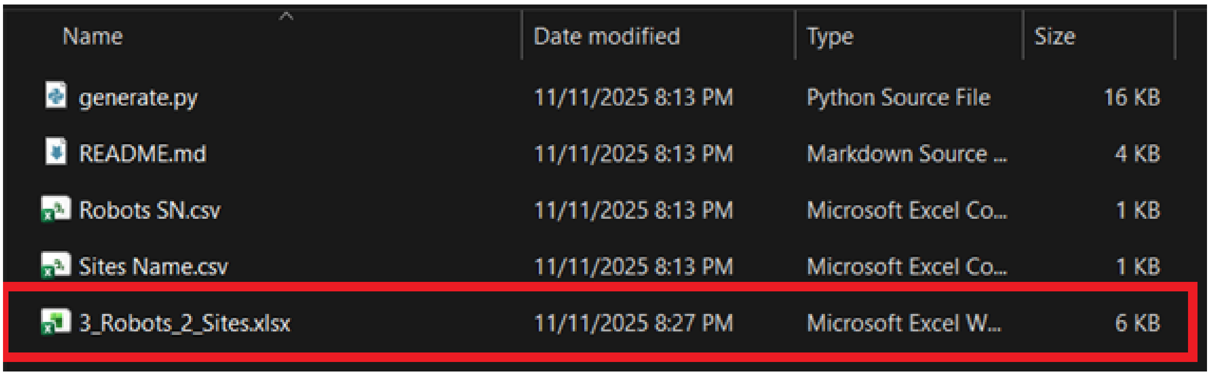


- Then write this command “python generate.py” then click “Enter”



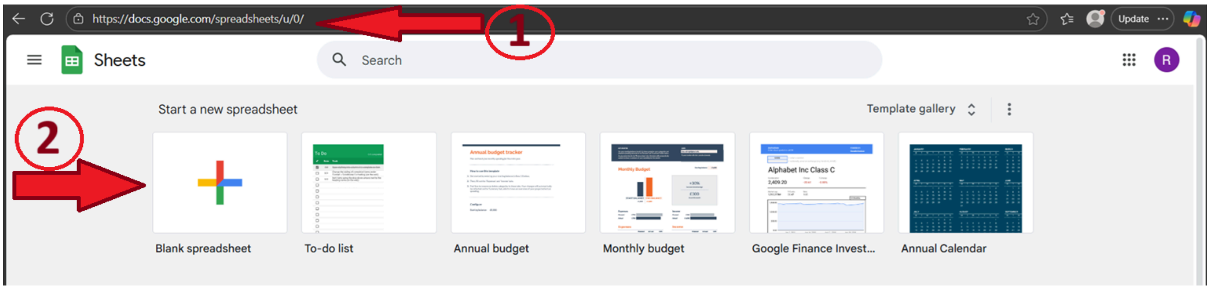
Finally: The Output Excel File was generated 🥳🤩

It’s name will be according to #number of Robots and #number of Sites

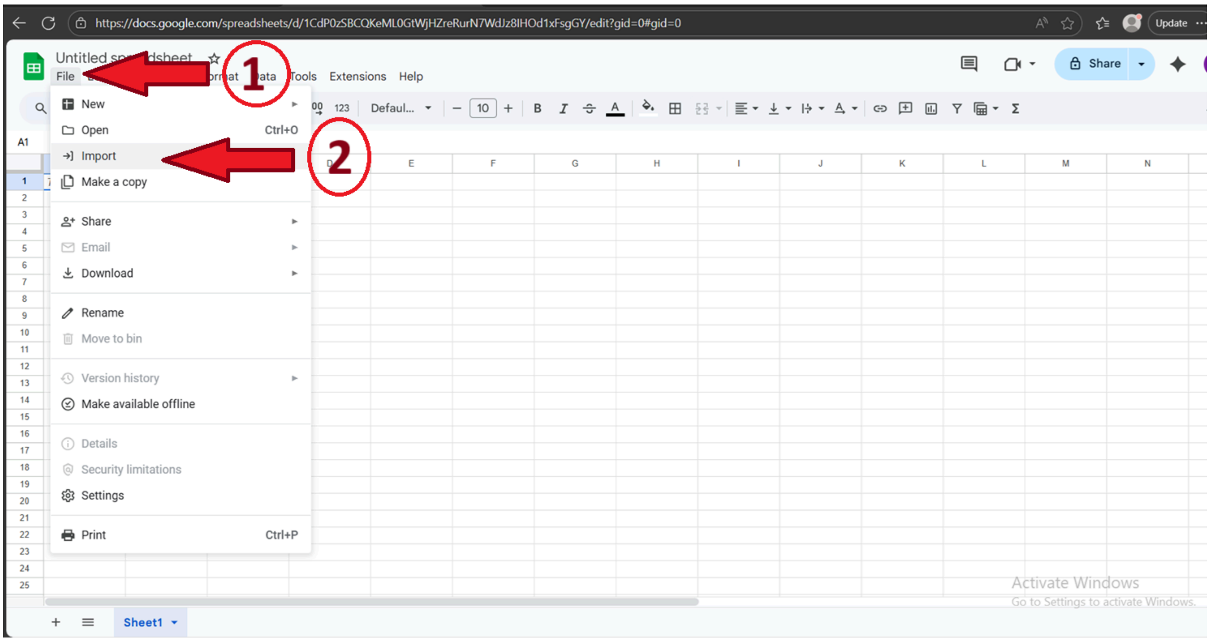


Step 5:

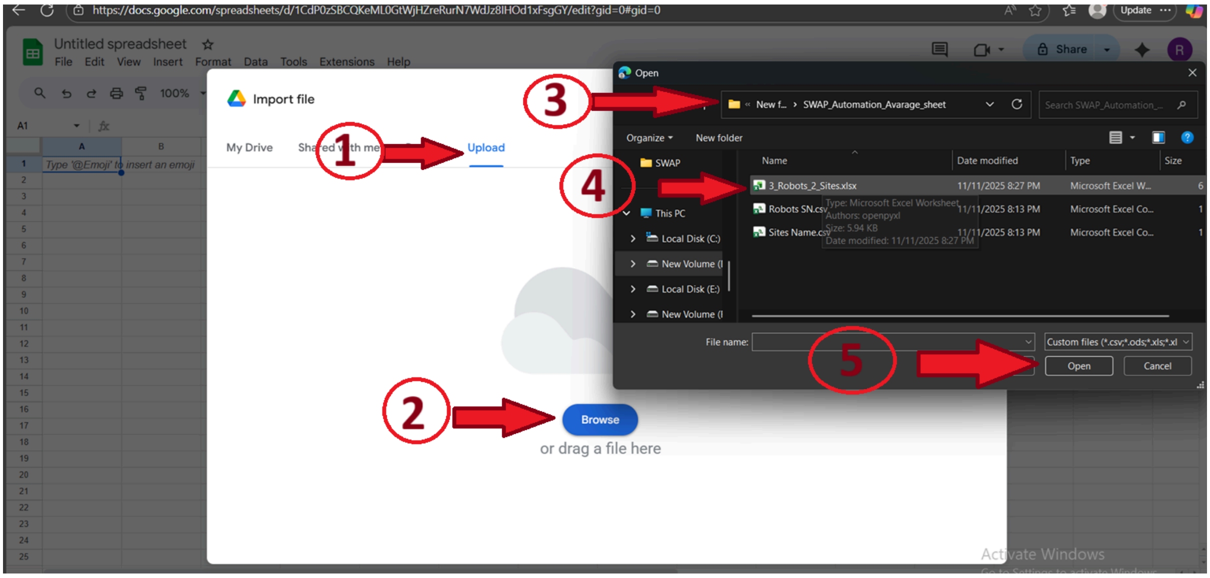
- Open https://docs.google.com/spreadsheets
  -open a blank spreadsheet

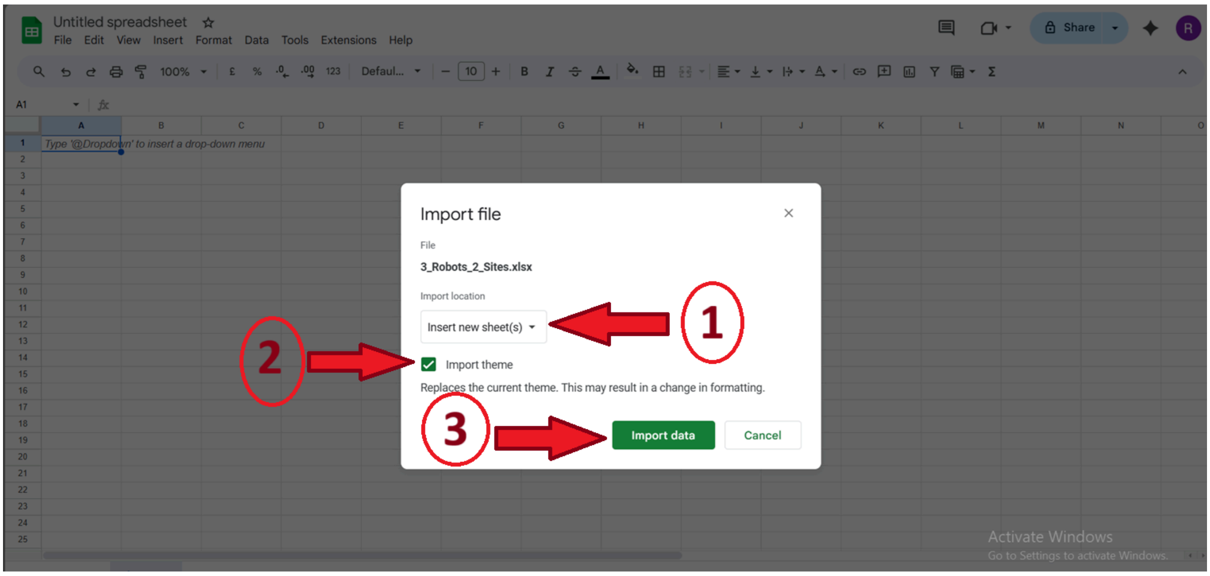


- The choose “file -> import”



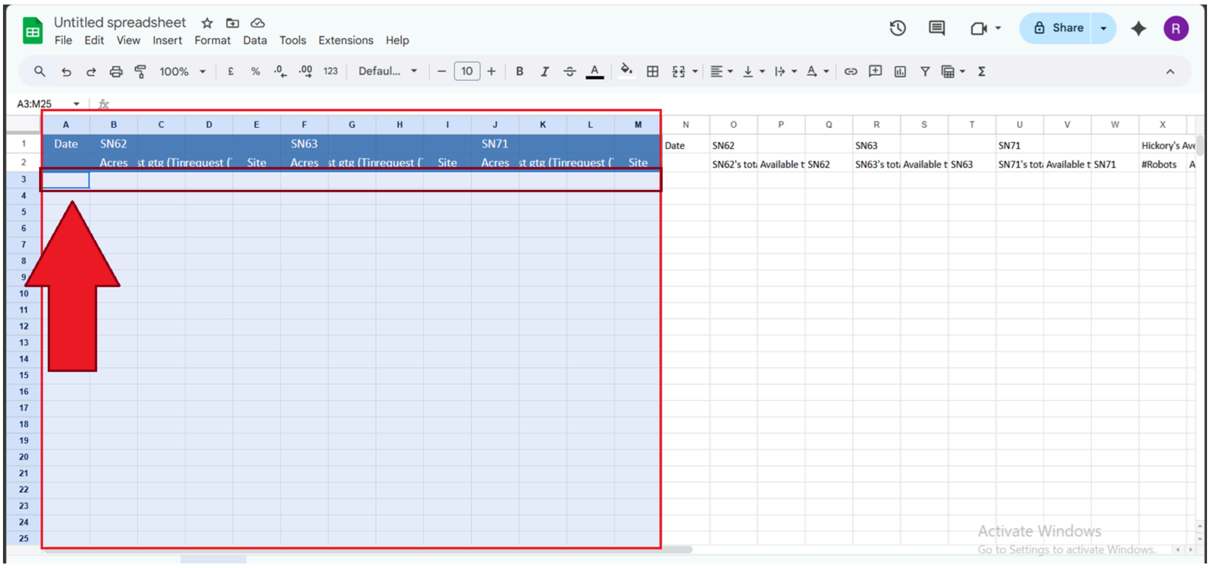
- Then upload the Output Excel File was generated.





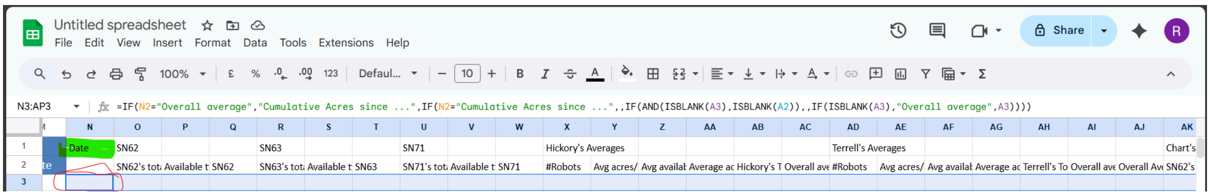
Step 6:

- Final finishes:
  - first you can fill your actual data in theses columns:
    - From Date in column (A) to final Robot site column

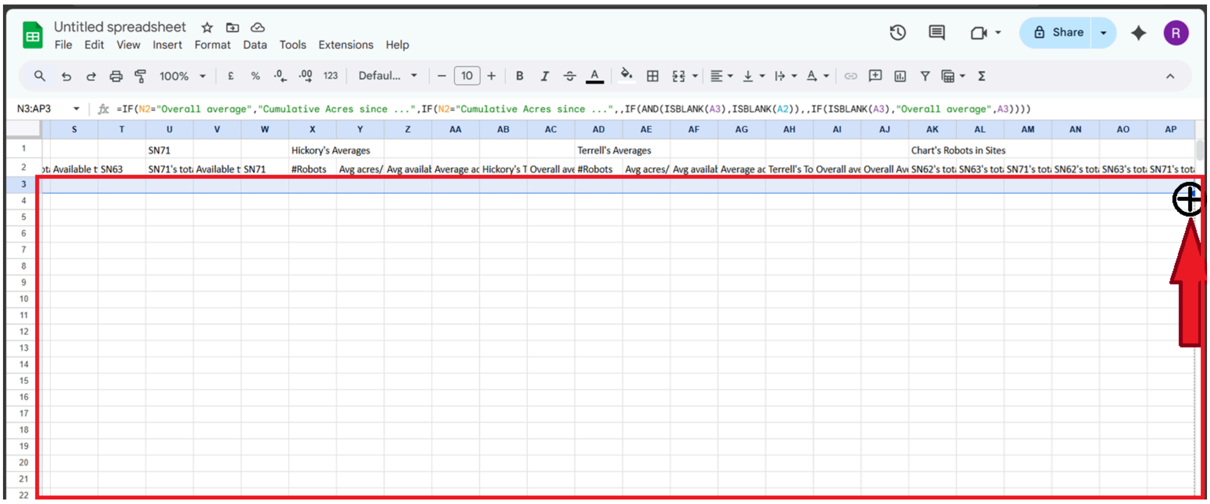


- Now you should pull the equations in the Row 3 to fill Automatically the remains Rows

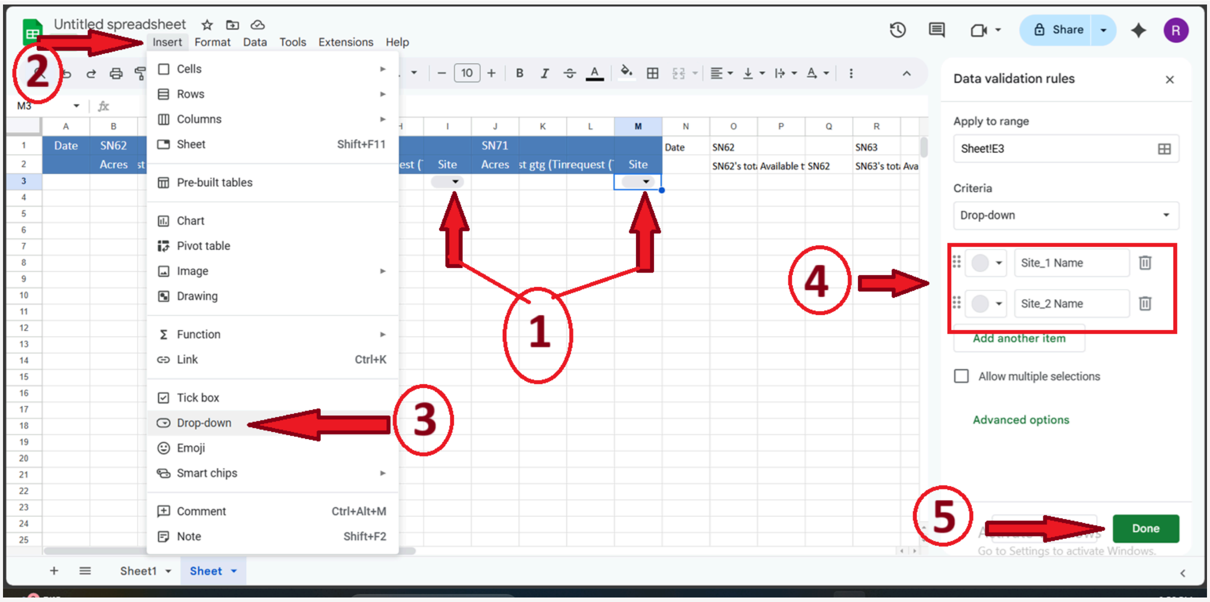
Select these cells from *the start * witch the second Date column



To the last column - then pull the Cross pointer mouse to the last Rows you want

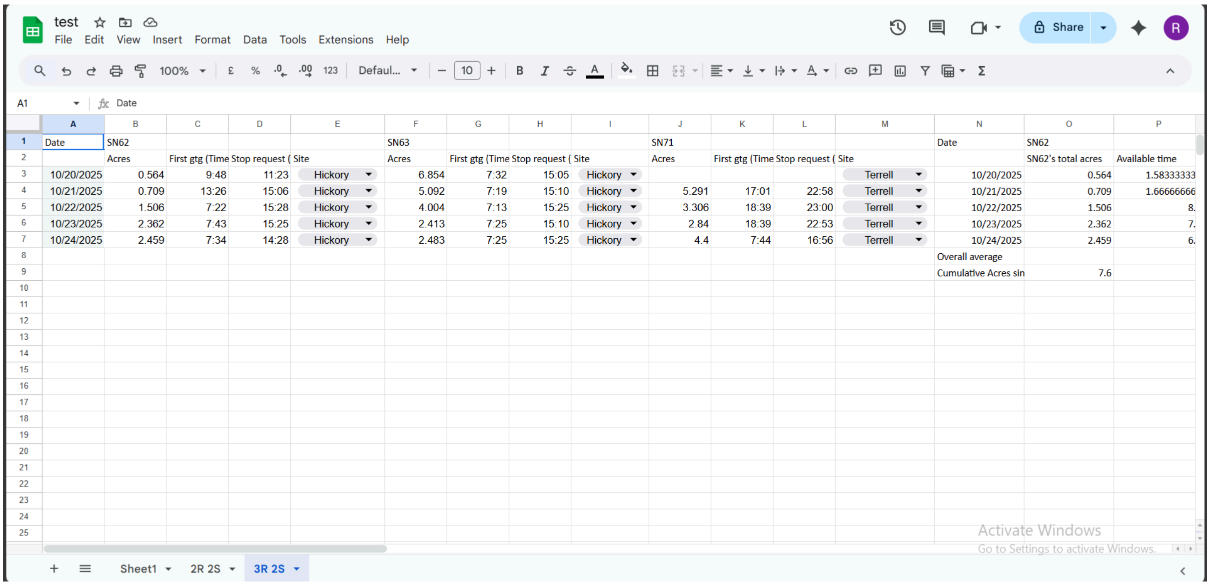


- Now finally fill the Dropdawn list of Robot's Sites with the Data of in `Sites Name.csv`



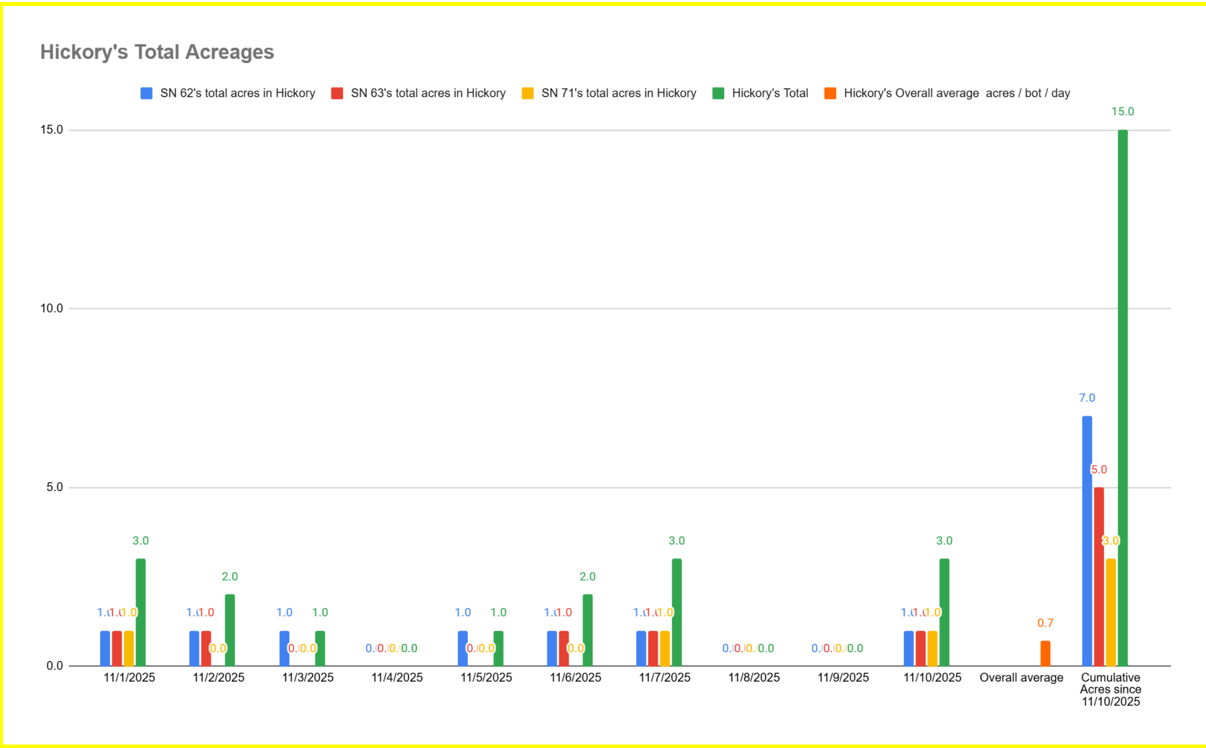
Step 7:

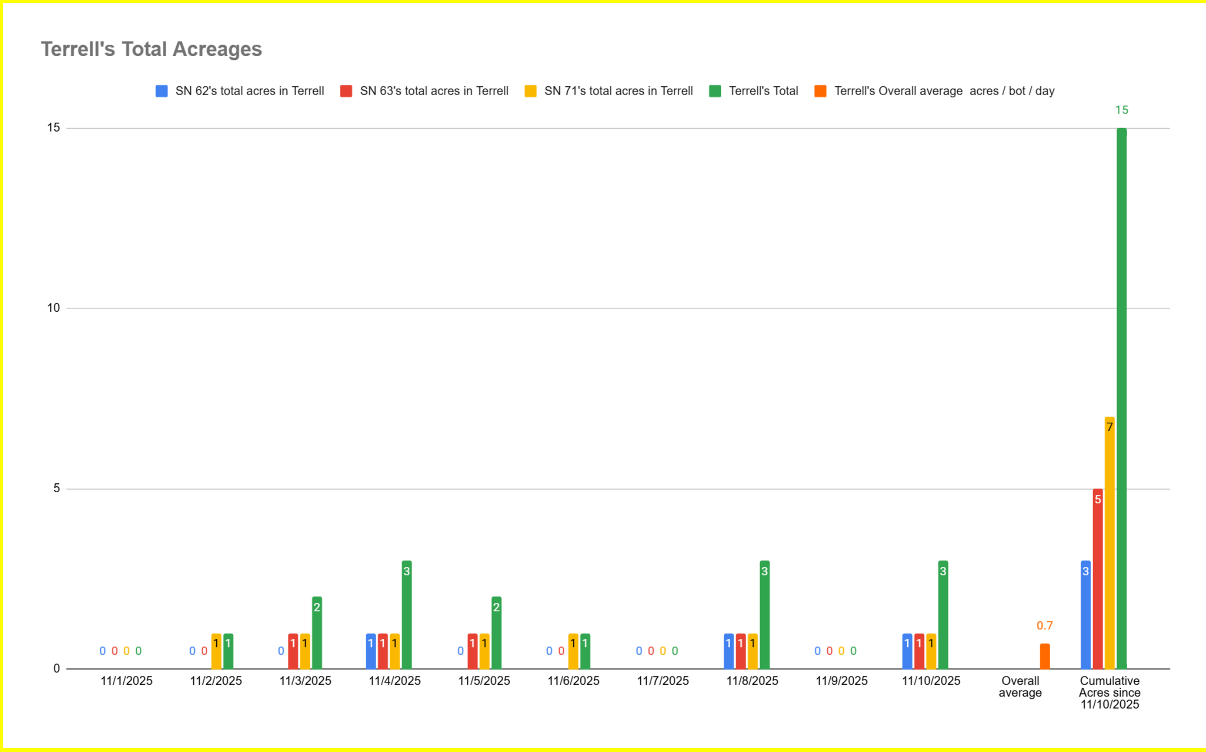
- Now you can fill your actual date and see the summery



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Date	SN62				SN63				SN71				Date	SN62	
2		Acres	First gtg (Time	Stop request (	Site	Acres	First gtg (Time	Stop request (	Site	Acres	First gtg (Time	Stop request (	Site		SN62's total acres	Available time
3	10/20/2025	0.564	9:48	11:23	Hickory	6.854	7:32	15:05	Hickory				Terrell	10/20/2025	0.564	1.58333333
4	10/21/2025	0.709	13:26	15:06	Hickory	5.092	7:19	15:10	Hickory	5.291	17:01	22:58	Terrell	10/21/2025	0.709	1.66666666
5	10/22/2025	1.506	7:22	15:28	Hickory	4.004	7:13	15:25	Hickory	3.306	18:39	23:00	Terrell	10/22/2025	1.506	8.
6	10/23/2025	2.362	7:43	15:25	Hickory	2.413	7:25	15:10	Hickory	2.84	18:39	22:53	Terrell	10/23/2025	2.362	7.
7	10/24/2025	2.459	7:34	14:28	Hickory	2.483	7:25	15:25	Hickory	4.4	7:44	16:56	Terrell	10/24/2025	2.459	6.
8														Overall average		
9														Cumulative Acres sin	7.6	

- **<u>Finally:</u>** You can generate your Desired Charts 🥳🤩🥳🤩🥳🤩🥳🤩🥳🤩🥳🤩🥳🤩🥳




Terrell's Total Acreages
SN 62's total acres in Terrell
SN 63's total acres in Terrell
SN 71's total acres in Terrell
Terrell's Total
Terrell's Overall average acres / bot / day
15
10
5
0
0 0 0 0
0 0 1 1
0 1 1 2
1 1 1 3
0 1 1 2
0 0 1 1
0 0 0 0
1 1 1 3
0 0 0 0
1 1 1 3
0.7
3 5 7 15
11/1/2025
11/2/2025
11/3/2025
11/4/2025
11/5/2025
11/6/2025
11/7/2025
11/8/2025
11/9/2025
11/10/2025
Overall average
Cumulative Acres since 11/10/2025